Seat No: _____________ Enrollment No: ___________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech Summer 2025 - 26 Examination

Semester: 8th **Date: 15/04/2026**
Subject Code: 203105483 **Time: 10:30 am to 1:00 pm**
Subject Name: Data Science **Total Marks: 60**

Instructions:
1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1 is compulsory,** From Section B, **Q.1 is compulsory.**
3. Figures to the right indicate full marks.
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

	SECTION-A	Marks	CO	BT
Q.1	Answer the following questions.			
	A.	[06]		
	1. Define types of data with examples.		1	2
	2. Explain sources of data.		2	2
	3. Explain data cleaning process.		2	3
	B.	[06]		
	1. Define central tendency (mean, median, mode).		3	2
	2. Describe Naive Bayes algorithm.		3	3
	3. Explain types of data visualization.		4	2
Q.2	A. Explain the complete data science process with a real-world example.	[04]	1	3
	B. Discuss types of data (structured, unstructured, semi-structured) and their importance in data science.	[05]	1	4
	OR			
	B. Explain data collection methods from multiple sources and challenges in data cleaning.	[05]	2	4
Q.3	A. Describe data storage and management techniques in data science.	[04]	2	3
	B. Explain statistical concepts: central tendency, variance, and distribution with examples.	[05]	3	4
	OR			
	B. Compare machine learning algorithms: Linear Regression, SVM, and Naive Bayes.	[05]	3	4
	SECTION-B			
Q.1	Answer the following questions.			
	A.	[06]		
	1. Explain visual encoding.		4	2
	2. Explain Bokeh for data visualization.		5	2
	3. Describe applications of data science.		5	2
	B.	[06]		
	1. Explain recent trends in data science.		6	2
	2. Describe modern visualization techniques.		4	2
	3. Explain data science toolkit.		1	2
Q.2	A. Explain sampling techniques and Central Limit Theorem (CLT) with diagram.	[04]	3	3

	B. Explain data visualization principles, including retinal variables and visual encoding.	**[05]**	**4**	**4**
	OR			
	B. Discuss types of data visualization and their appropriate use cases.	**[05]**	**4**	**4**
Q.3	**A. Design a data visualization strategy for a given dataset (steps and example).**	**[04]**	**4**	**4**
	B. Explain applications of data science in real-world scenarios (healthcare, finance, e-commerce).	**[05]**	**5**	**3**
	OR			
	B. Explain recent trends in data science, including modern visualization techniques and application development methods.	**[05]**	**6**	**4**